

SCOTT AFB, IL, USA

WMO: 724338

Lat: 38.550N

Lon: 89.850W

Elev: 140

StdP: 99.66

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.1	-11.3	-19.0	0.7	-12.7	-16.2	0.9	-10.0	11.9	1.5	10.7	0.7	4.1	330	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	35.0	25.4	33.7	25.0	32.4	24.5	27.1	32.0	26.3	31.3	25.5	30.4	3.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	21.6	29.6	25.0	20.4	29.0	24.1	19.3	28.1	86.1	32.0	82.3	31.0	79.0	30.4	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	7.8	-18.6	37.7	3.2	3.0	-20.9	39.8	-22.8	41.5	-24.6	43.2	-26.9	45.4		
(5)			-18.6	28.5	3.4	0.9	-21.1	29.2	-23.1	29.7	-25.0	30.2	-27.5	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.5	-0.6	2.6	7.6	13.6	19.1	23.9	25.6	25.0	20.8	14.6	7.4	2.0
	DBStd	10.33	6.51	6.09	6.03	5.06	4.36	3.29	2.95	2.85	3.94	4.90	5.24	5.62
	HDD10.0	1070	333	217	119	22	1	0	0	0	0	13	110	255
	HDD18.3	2561	587	439	336	157	46	3	0	1	19	137	328	508
	CDD10.0	2354	5	11	43	129	283	417	485	464	324	156	32	6
	CDD18.3	803	0	0	2	13	69	170	227	206	94	21	1	0
	CDH23.3	7929	0	2	27	140	599	1661	2396	1986	903	207	9	0
	CDH26.7	3257	0	0	2	27	177	695	1094	862	346	52	1	0
Wind	WSAvg	3.3	4.0	4.0	4.1	4.1	3.2	2.8	2.4	2.2	2.3	3.0	3.5	3.7
Precipitation	PrecAvg	1112	66	58	85	121	132	121	100	94	87	81	93	73
	PrecMax	1441	208	121	254	288	380	244	185	163	236	290	181	206
	PrecMin	792	11	16	26	46	19	31	16	34	11	15	20	18
	PrecStd	165	42	28	44	59	81	62	46	35	55	54	48	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	22.2	26.3	29.8	32.3	35.5	37.5	36.6	34.6	30.8	24.8	19.0
		MCWB	15.3	15.4	17.8	19.8	22.8	24.8	25.3	25.5	23.1	21.8	17.1	16.1
	2%	DB	15.3	18.2	23.0	27.2	30.4	33.8	35.1	34.7	32.6	27.9	21.1	16.2
		MCWB	12.2	13.9	15.7	18.6	22.2	24.5	26.0	25.5	23.4	19.3	15.2	13.9
	5%	DB	12.4	15.5	20.3	24.8	28.8	32.4	33.7	32.8	30.8	25.7	18.5	13.5
		MCWB	10.0	11.7	14.4	17.5	21.4	24.2	25.6	25.1	22.6	18.5	14.5	10.9
	10%	DB	8.8	12.4	17.5	22.5	27.0	30.9	32.2	31.3	28.6	23.0	16.2	10.3
		MCWB	6.4	9.1	12.9	16.4	20.5	23.5	25.0	24.2	21.1	17.3	12.7	8.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.2	17.2	19.3	21.8	25.2	27.1	28.4	28.2	25.8	23.3	19.1	17.2
		MCDB	17.9	19.5	24.0	27.0	29.4	31.9	33.0	32.7	31.2	28.5	21.7	18.6
	2%	WB	13.1	15.0	17.4	20.3	23.6	26.0	27.3	27.0	24.8	21.6	17.2	14.5
		MCDB	14.5	17.7	21.1	24.6	28.2	31.3	32.3	32.0	29.8	25.6	19.1	15.8
	5%	WB	9.8	12.2	15.7	18.9	22.6	25.2	26.5	26.0	23.8	20.1	15.4	11.4
		MCDB	11.7	14.6	19.0	22.8	26.8	30.3	31.5	30.9	28.1	23.4	18.2	12.8
	10%	WB	6.0	8.8	13.7	17.6	21.6	24.4	25.7	25.2	22.7	18.6	13.0	8.2
		MCDB	8.4	12.2	17.1	21.0	25.5	29.1	30.5	29.8	26.6	22.4	15.7	9.8
Mean Daily Temperature Range	5% DB	MDBR	9.3	10.0	11.3	11.9	11.1	11.0	10.9	11.3	12.7	12.6	10.7	9.0
		MCDBR	13.7	15.0	15.2	14.5	12.8	12.8	12.3	12.9	14.5	15.1	14.3	12.9
		MCWBR	10.5	10.7	8.9	8.0	6.3	5.1	4.8	5.3	6.0	7.6	9.4	10.1
	5% WB	MCDBR	12.4	13.7	12.9	12.1	11.0	11.1	11.1	11.5	11.8	12.2	11.8	11.2
		MCWBR	10.6	11.0	8.6	7.8	6.2	5.2	5.1	5.4	5.5	7.4	9.4	10.1
Clear-Sky Solar Irradiance	taub	0.308	0.318	0.352	0.403	0.437	0.444	0.472	0.465	0.410	0.353	0.313	0.302	
	taud	2.466	2.424	2.394	2.248	2.212	2.223	2.147	2.160	2.296	2.460	2.513	2.497	
	Ebn at Noon	872	908	907	875	848	839	812	808	834	858	860	853	
	Edh at Noon	81	96	109	134	142	141	151	145	120	92	76	73	
All-Sky Solar Radiation	RadAvg	2.04	2.76	3.74	4.95	5.55	6.38	6.22	5.61	4.73	3.43	2.32	1.71	
	RadStd	0.22	0.29	0.36	0.29	0.49	0.44	0.43	0.32	0.31	0.37	0.25	0.18	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+78

Nomenclature: See separate page